

Set 5: Website Analytics

```
install.packages("ggplot2")
install.packages("gridExtra")
library(ggplot2)
library(gridExtra)

inventory <- data.frame(
  ProductID = c(1, 2, 3, 4, 5),
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  QuantityAvailable = c(250, 175, 300, 200, 220),
  ProductPrice = c(500, 350, 700, 450, 600),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories")
)

print(inventory)

bar_chart <- ggplot(inventory,
  aes(x = ProductName,
    y = QuantityAvailable,
    fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(title = "Quantity Available by Product",
    x = "Product Name",
    y = "Quantity Available") +
  theme_minimal() +
  theme(legend.position = "none")

print(bar_chart)

stacked_chart <- ggplot(inventory,
  aes(x = Category,
    y = QuantityAvailable,
    fill = ProductName)) +
  geom_bar(stat = "identity") +
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labs(title = "Quantity Available by Product Category",
      x = "Category",
      y = "Quantity Available") +
theme_minimal()

print(stacked_chart)

scatter_plot <- ggplot(inventory,
                       aes(x = ProductPrice,
                           y = QuantityAvailable,
                           color = ProductName)) +
geom_point(size = 4) +
geom_text(aes(label = ProductName),
          vjust = -1,
          show.legend = FALSE) +
labs(title = "Product Price vs Quantity Available",
      x = "Product Price",
      y = "Quantity Available") +
theme_minimal()

print(scatter_plot)
cat("\n-----\n")
cat("Findings:\n")
cat("-----\n")
cat("1. Product C has the highest inventory (300 units).\n")
cat("2. Product B has the lowest inventory (175 units).\n")
cat("3. Electronics category contains Products A and B.\n")
cat("4. Furniture category contains Products C and D.\n")
cat("5. Accessories category contains Product E.\n")
cat("6. Higher-priced products do not necessarily have lower inventory levels.\n")
grid.arrange(
  bar_chart,

```

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stacked_chart,

scatter_plot,

ncol = 2,

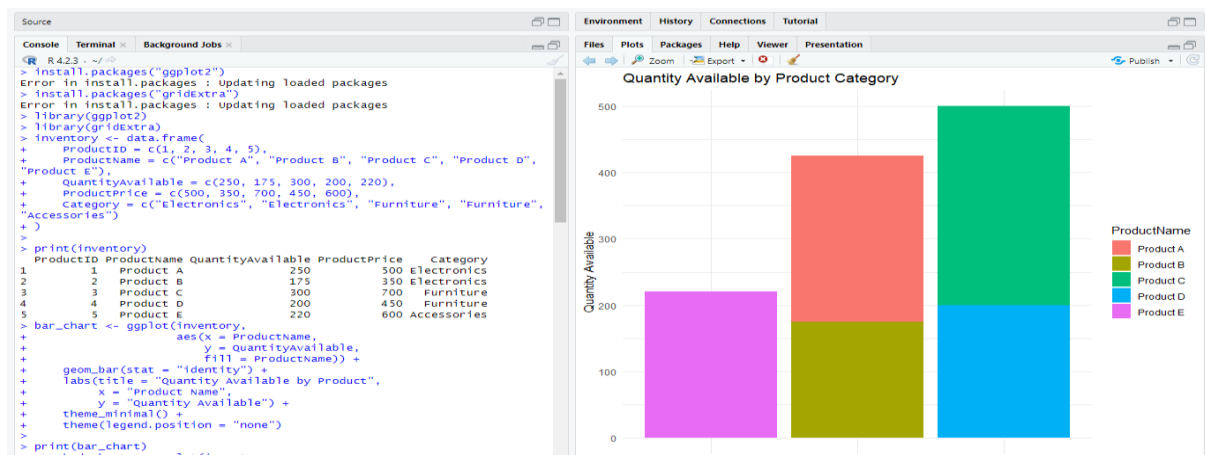
top = "PRODUCT INVENTORY DASHBOARD"
)

```

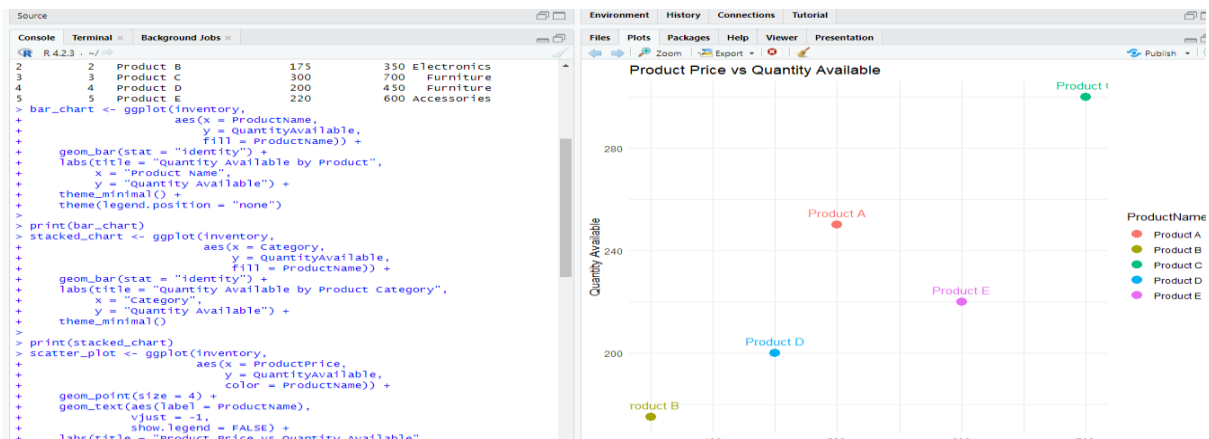
1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.



3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.



4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

